

A Virtual Peer Mentor to Enhance Social Presence in VR Rehabilitation for Recovering Heart-Attack Patients

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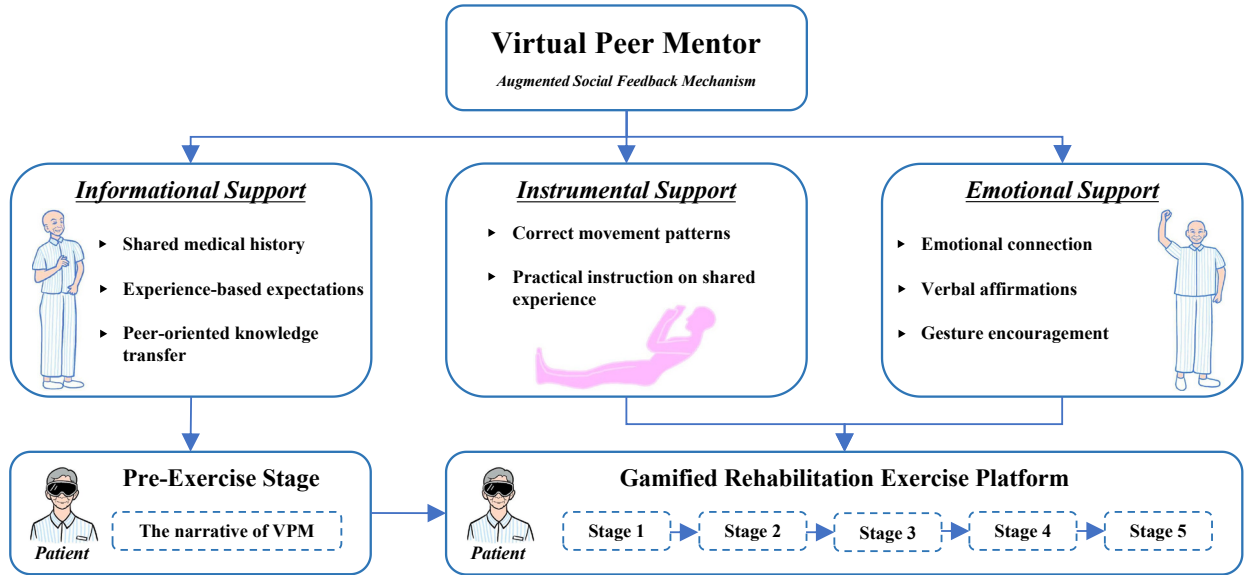


Fig. 1: Overview of the system architecture of the cardiac rehabilitation platform, featuring a custom immersive virtual-reality rehabilitation game integrated with virtual patient mentor (VPM) and three social support modules, informational, instrumental and emotional, grounded in social support theory.

Abstract— The adoption of immersive virtual reality (IVR) for gamified rehabilitation is increasing. However, a significant challenge relates to perceptions of virtual environments as empty and isolating, potentially increasing stress, particularly among older users (patients). This paper explores the use of a virtual peer mentor (VPM) in a custom IVR-rehabilitation game to provide guidance and companionship. This game specifically targets patients recovering from acute myocardial infarction (AMI), commonly known as a heart attack. Grounded in social support theory, the VPM provides three types of support: (1) informational support, through pre-exercise narratives detailing a shared medical history; (2) instrumental support, through real-time demonstrations of clinically-validated exercise movements; and (3) emotional support, through positive feedback and encouragement. A within-subjects study involving 30 hospitalized AMI patients (all over 47 years old) evaluated the effectiveness of the VPM-integrated IVR-rehabilitation game. Each participant experienced a baseline (no VPM) and VPM-integrated version of the game on separate days. The results from the Intrinsic Motivation Inventory (IMI) and the social presence module of the Game Experience Questionnaire (GEQ-SPM) show that the VPM resulted in significant increases in engagement, and statistically significant lower pressure/tension. Furthermore, participants exhibited high user acceptance (76.7%) and task-completion rates (98.5%), with minimal cybersickness. The findings demonstrate that a psychologically-grounded VPM can effectively reduce stress in middle- and older-age patients in an IVR rehabilitation setting.

Index Terms—Immersive virtual reality, virtual peer mentor, rehabilitation, serious game, acute myocardial infarction

1 INTRODUCTION

Immersive virtual reality (IVR)-based rehabilitation systems often integrate game-oriented approaches, such as gamification [33], serious games [27], and game-based learning [10], to enhance patient engagement. In recent years, gamifying IVR has become a widespread method that applies game design features to rehabilitation protocols while maintaining core health objectives [15]. This approach transforms repetitive

and tedious exercises into highly engaging, intrinsically motivating experiences through integration of the reward mechanism, progress tracking, and interactive elements aligned with rehabilitation goals [33].

Empirical studies have examined various feedback mechanisms in gamified IVR systems to support rehabilitation [58], providing immediate performance feedback that maintains motivation, corrects inappropriate movements, and reinforces positive behaviors [14]. Common real-time modalities include visual feedback, such as progress indicators and movement-responsive animations [60], auditory signals, such as encouraging sounds or corrective signals [35], and haptic feedback, such as vibrations that indicate task completion or errors [30]. To improve feedback delivery, many studies integrate virtual avatars as interactive agents that provide dynamic feedback [3, 56]. These avatars can act as virtual instructors that demonstrate exercises [47] or as personalized guides in scenario-based training [16], which promotes a more engaging and responsive rehabilitation experience.

Although gamified IVR-based rehabilitation systems have been

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Received 12 September 2025; revised 16 January 2026; accepted 23 January 2026.

Date of publication 30 March 2026; date of current version 12 May 2026.

This article has supplementary downloadable material available at <https://doi.org/10.1109/TVCG.2026.3679134>, provided by the authors.

Digital Object Identifier no. 10.1109/TVCG.2026.3679134

shown to improve participation, they can still be experienced as empty and isolating socially, potentially increasing stress among middle-aged and older patients with acute myocardial infarction (AMI) [64]. Within cardiac rehabilitation, psychosocial factors are widely recognized as clinically relevant and are related to adherence and lifestyle outcomes [4]. Practical and social support predicts the attendance of cardiac rehabilitation [32]. Large-scale evidence further links loneliness to a higher risk of AMI events [13]. Importantly, lonely individuals have a 61% higher risk of a 30-day emergency department visit or readmission [1]. Together, these findings suggest a specific mechanism relevant to immersive rehabilitation: when rehabilitation is experienced without a social companion, the perceived social presence may reduce active engagement. However, previous work on IVR-based rehabilitation has largely emphasized superficial gamification or instructor-like avatars rather than psychologically grounded, peer-oriented designs that explicitly target social presence as a stress-alleviating mechanism [53]. In addition, many evaluations are based on healthy volunteers, which may not reflect the physical limitations and emotional challenges of actual AMI patients [5, 11]. Consequently, a critical research gap remains in utilizing socially supportive virtual peers to address the psychological needs of AMI patients during cardiac rehabilitation.

Middle-aged adults (45–60 years) and older adults (over 60 years) constitute a significant segment of the population of patients with AMI [23]. However, the acceptance and experiential outcomes of these demographics in IVR-assisted rehabilitation remain underexplored [2]. Schaumburg *et al.* [46] identify several age-related barriers to IVR participation. These include physical restrictions (e.g. reduced agility, sensory decline), cognitive changes (e.g. executive dysfunction, memory, and attention deficits), lower technological proficiency, and increased skepticism. Together, these factors increase the risk of IVR-related complications, such as cybersickness, psychological distress, and physical fatigue. Consequently, understanding IVR acceptance and performance within virtual environments is essential for effective clinical integration and therapeutic efficacy.

To address these research gaps, this study applies the social support theory [50] to design a virtual peer mentor (VPM) that serves as an enhanced feedback mechanism. Designed for middle-aged and older patients with AMI, the VPM simulates a shared rehabilitation journey to target their psychological needs and alleviate stress during IVR-based therapy. It delivers three forms of support: (1) informational support through pre-exercise narratives that convey a shared medical history; (2) instrumental support via real-time demonstrations of clinically validated exercise movements; and (3) emotional support by providing positive feedback and encouragement. This study aims to answer the following two research questions (RQs):

RQ1: Does the integration of a VPM into the feedback mechanism reduce short-term pressure and tension while increasing behavioral involvement among middle-aged and older AMI patients in IVR-assisted rehabilitation?

RQ2: What key factors influence the acceptance and experience of IVR-assisted rehabilitation among middle-aged and older AMI patients?

This study makes two primary contributions. First, it introduces a VPM design grounded in social support theory. Integrated within an IVR game, the VPM provides identity-matched companionship and feedback, to reduce patient stress and improve engagement. Second, it presents a comprehensive mixed-methods evaluation to identify the key factors influencing how middle-aged and older patients with AMI accept and experience IVR-assisted rehabilitation. Using an identity-matched VPM, rather than a generic virtual instructor or authority figure [53], this work advances IVR-based cardiac rehabilitation by demonstrating how socially responsive virtual peers can effectively address the specific psychosocial needs of this demographic.

2 RELATED WORK

Recent studies have extensively investigated visual feedback mechanisms within IVR-assisted rehabilitation [44]. For example, Winter *et al.* [60] used a progress bar to show patients with multiple sclerosis

their improvements in walking speed, increasing engagement through visible goal progression and a sense of achievement. Similarly, Kearney *et al.* [20] found that augmented visual feedback, which translates patient movements into immediate visual cues about performance quality, facilitates self-correction, strengthens self-efficiency, and maintains motivation during repetitive training. Compared to conventional approaches that lack timely visualization, these feedback mechanisms offer clearer guidance and a more intrinsically rewarding rehabilitation experience.

Beyond visual cues, other studies highlight the benefits of auditory and haptic feedback. Ghai *et al.* [11] demonstrated that rhythmic auditory signals facilitate motor consistency and participation in repetitive exercises. Chen *et al.* [7] further showed that auditory kinematic feedback, which maps real-time audio signals to movement velocity or joint angles, allows patients to immediately perceive and correct their performance. Haptic feedback has also been proven to be effective. Camardella *et al.* [5] used haptic feedback as a source of sensory information for upper limb rehabilitation in children with cerebral palsy, demonstrating that it reduces speed tracking errors and increases movement fluidity. Zhang *et al.* [63] found that haptic feedback significantly improves user embodiment and virtual interactive experience compared to no stimulation, particularly among patient populations. Although these studies thoroughly examine external feedback that uses visual, auditory, and haptic cues, they largely overlook internal feedback processes. Key elements such as self-assessment, emotional fulfillment, personal accomplishment, and intrinsic motivation are often neglected. These internal factors are essential for sustained engagement and meaningful behavioral change, representing a significant gap in current research.

Meanwhile, integrating virtual avatars into IVR rehabilitation has appeared as a promising strategy to personalize patient engagement and improve therapeutic outcomes [53]. For instance, Segear *et al.* [47] designed a virtual coach to guide users through lower-extremity balance rehabilitation, reporting significant performance gains when the avatar provided verbal instructions, while a virtual health agent developed by Hartanto *et al.* [17] successfully reduced anxiety in patients with social phobia. Mihajlovic *et al.* [31] showed that a gamified neck-rehabilitation task using a non-humanoid butterfly target that generates rapid head-neck movements, smooth tracking, and static holds was perceived as intuitively motivating and playful, while also promoting adherence to prescribed movements, maintaining engagement, and providing objective progress feedback.

However, many existing IVR rehabilitation systems that deploy avatars serve primarily as instructional agents and do not treat social presence as a core, psychosocially grounded design objective [53]. This concern is particularly relevant in cardiac rehabilitation, where psychosocial factors influence whether patients remain engaged in recommended recovery behaviors [4]. Pogosova *et al.* [42] highlighted that psychosocial risk factors, including social isolation, low social support, stress, depression, and anxiety, can reduce adherence to rehabilitation. In the context of heart failure exercise training, Cooper *et al.* [9] reported that psychosocial and participation barriers are associated with adherence to exercise. Similarly, Yang *et al.* [61] found that greater perceived social support is associated with greater adherence to prescribed rehabilitation exercises among patients with chronic heart failure. Together, these findings suggest that companionship and perceived support can help maintain engagement and adherence to rehabilitation. Nevertheless, within IVR rehabilitation, few studies use avatars to provide companionship and social presence or evaluate how these features influence engagement, stress, and adherence during rehabilitation.

Several empirical studies have explored the effectiveness of IVR-assisted rehabilitation in broad age ranges [18]. For example, Pereira *et al.* [39], in a study with participants aged 5 to 85, identify the key advantages of IVR for hand rehabilitation with participants. They found that IVR enhances motivation through engaging exercises, real-time feedback, and immersive experiences. IVR also allows individualized progression through adaptive difficulty and enables objective performance tracking for clinical evaluation, making it a feasible and scalable

option for home care. Focusing on middle-aged and older adults, a systematic review by Maier *et al.* [26] analyzed the effects of IVR on post-stroke motor recovery. This review concluded that custom-made IVR systems designed around rehabilitation principles, including task-specific practice (functionally meaningful movements relevant for daily living), variable practice (diverse tasks requiring different movements), implicit feedback (real-time visualization of kinematic movement data), explicit feedback (information about task completion) and increasing difficulty (adaptive progression based on patient ability), may deliver effective post-stroke motor rehabilitation compared to conventional therapy.

Many IVR studies use healthy individuals as participants, and their findings consistently highlight the importance of considering age in rehabilitation design. However, as noted by Peck *et al.* [36], the field suffers from a lack of participant diversity, often relying on samples that do not reflect the target clinical population. For instance, Shah *et al.* [48] reported a case study involving healthy elderly players aged 60 to 88 years in a collaborative IVR-based five-week game intervention. Their findings showed that social aspects are key motivating factors, suggesting that older adults may have increased social connection needs in technology-mediated rehabilitation environments. Similarly, Chen *et al.* [8], studying participants aged 20 to 79 years, concluded that age significantly influences the acceptance of IVR games for upper-limb exercises, reinforcing the need for age-specific strategies. Although these studies provide valuable information, their reliance on healthy participants is a key limitation. Ziadeh *et al.* [65] acknowledge this, noting that the findings of healthy young adults (aged 20 to 28 years) who completed IVR-based motor imagery tasks, for example, require future validation in actual stroke patients. This is a critical point, as patients dealing with physical restrictions, pain, and psychological stress can perceive virtual experiences such as agency and embodiment much differently than their healthy counterparts.

Despite promising developments in IVR-assisted rehabilitation, research has three critical limitations. First, current feedback mechanisms are based on external drivers rather than addressing the internal psychological processes that sustain engagement. Second, virtual avatar designs typically act as instructors, lacking the social and peer-based support essential for the emotional journey of rehabilitation. Third, many studies are based on healthy individuals, which may not adequately represent the complex needs and experiences of actual patients.

To address these gaps, this study goes beyond purely cognitive stress models (e.g., Lazarus's cognitive appraisal theory [24]), which can be insufficient for managing the distress of a sudden and life-changing illness. Instead, we draw on social support theory [50], which posits that peer connection is a powerful buffer against stress. Since this social support is often absent during solitary rehabilitation, we developed a novel intervention to fill this crucial gap: the VPM. The VPM is an avatar modeled as a peer who has already recovered from a similar condition, offering shared experience, companionship, and encouragement. By integrating this social dimension of coping, our research aims to develop and assess a theoretically-informed IVR intervention that effectively reduces stress for patients with AMI during early cardiac rehabilitation.

3 METHODS

The intervention was developed as an IVR experience using a head-mounted display to utilize therapeutic mechanisms, including egocentric perception, tight sensorimotor coupling, and first-person embodiment, which are not fully realized by non-immersive screen-based VR experiences. The IVR-assisted VPM-integrated rehabilitation framework includes two main components: a VPM that serves as a social feedback mechanism and a gamified rehabilitation exercise platform. The VPM functions as the central social support intervention, included throughout a pre-exercise phase and five sequenced gamified rehabilitation stages to provide ongoing peer-based encouragement and companionship.

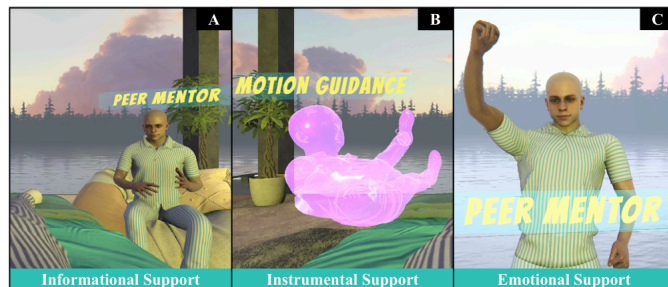


Fig. 2: The VPM delivers three forms of social support via corresponding modes: (A) informational support in narrative mode during the pre-exercise phase; (B) instrumental support in guidance mode, presented as a translucent pink demonstration of rehabilitation movements; and (C) emotional support in companion mode, in which the VPM remains beside the patient and provides encouragement through supportive phrases and gestures.

3.1 VPM Design

The VPM was conceptualized as an experientially matched virtual companion, modeled after individuals who had achieved functional recovery from a similar AMI. This study implemented a scripted framework rather than a complex conversational interface, and the interaction design was guided by the following considerations. First, for safety and auditability, fixed prompts minimize the risk of inappropriate content, facilitate clinical supervision, and ensure content compliance. Second, for reliability and standardization, consistent interactions among participants improve internal validity, allowing observed effects to be attributed to social presence rather than dialog variability. Third, for ease of clinical deployment, scripted content requires minimal computational resources and does not require continuous network connectivity, facilitating integration into hospital workflows. Finally, the design addresses the potential for dual-task interference in older adults, in which simultaneous cognitive and motor demands can affect performance [62].

In health-related contexts, avatar appearance influences perceived acceptability and persuasiveness [52], and avatar stylization can affect perceived trust [6]. Consequently, the appearance of the VPM was intentionally friendly and non-authoritative to ensure suitability for an inpatient clinical setting. Based on social support theory, the VPM was designed as a single peer-like agent comprising three supports: an informational support (narrative mode), an instrumental support (guidance mode) and an emotional support (companion mode) (see Fig. 2). Specifically:

- **Informational Support:** In the pre-exercise phase, the VPM appears in narrative mode as a fully rendered, human-like avatar with an energetic appearance. This design symbolizes a successful recovery state consistent with the narrative of the peer-mentor, providing a clear signal that recovery is achievable. In this mode, the VPM provides narrative content in three areas: (1) a shared medical history of AMI and successful recovery; (2) an overview of the five-stage rehabilitation process; and (3) evidence-based expectations for rehabilitation outcomes and timelines. This support promotes patient understanding and adaptation by facilitating peer-oriented knowledge transfer.
- **Instrumental Support:** Throughout the gamified exercises, the VPM appears in the guidance mode as a translucent pink silhouette. This visualization projects demonstrations of the movements used during the successful recovery of the peer mentor. The translucent rendering improves movement legibility and avoids occluding the game environment, signaling that the guidance is a supportive reference from a recovered peer rather than a co-present professional authority.
- **Emotional Support:** During exercise execution, the VPM appears in companion mode as a fully rendered human-like avatar and

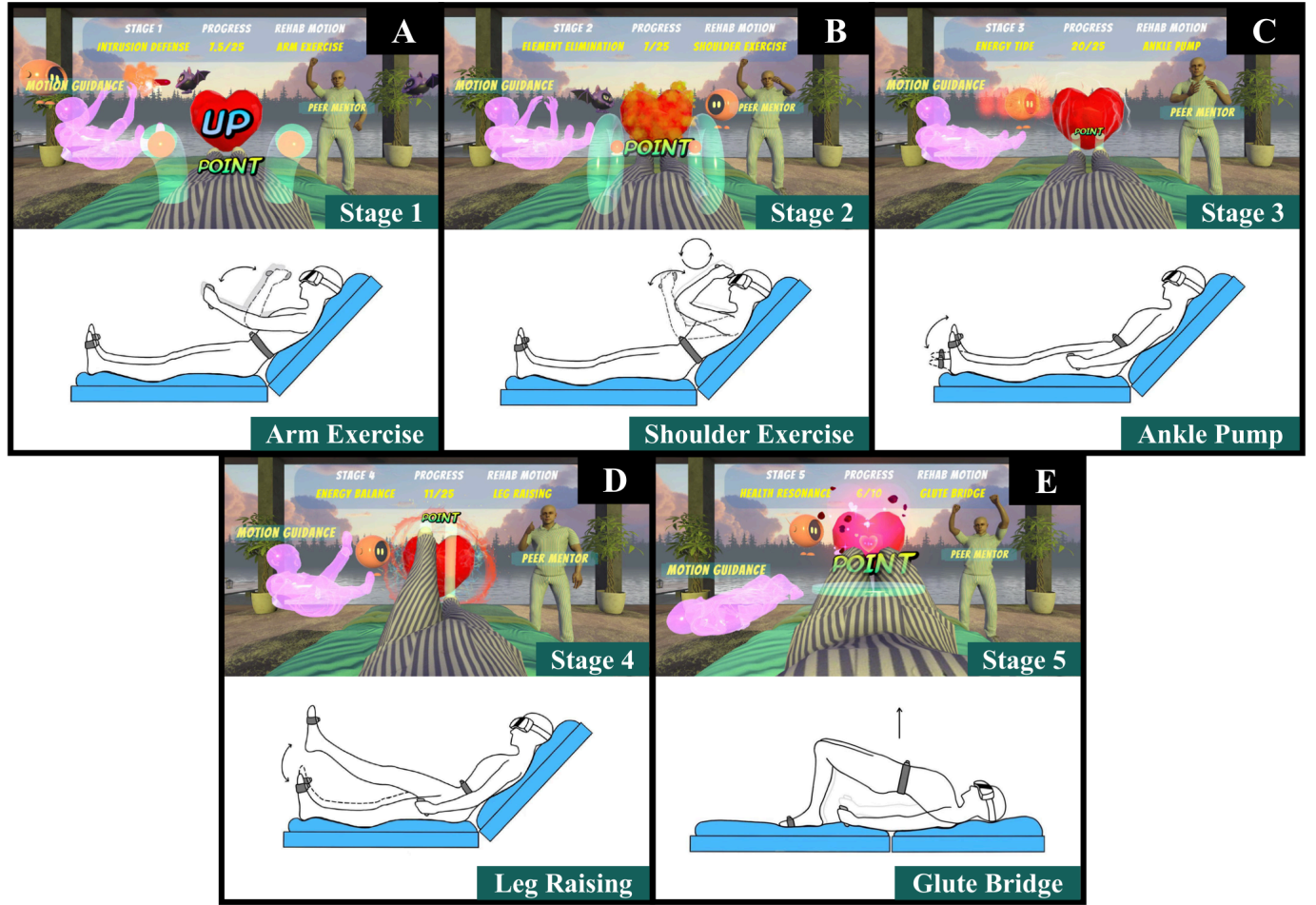


Fig. 3: Five stages of the IVR-assisted cardiac rehabilitation game for patients with AMI are as follows: (A) Stage 1 includes safeguarding a health crystal through arm exercises; (B) Stage 2 focuses on eliminating detrimental elements using shoulder exercises; (C) Stage 3 consists of injecting energy into a health crystal through ankle pump exercises; (D) Stage 4 involves stabilizing energy levels with leg raising exercises; and (E) Stage 5 involves assimilating health waves through glute bridge exercises.

remains with the patient as an encouraging presence. Upon successful completion of a movement, an encouraging phrase is randomly selected without repetition from a predefined library of 50 supportive statements adopted from [22], including expressions such “as well done” and “excellent work”. In parallel, the VPM performs three positive gestures (thumbs up, clapping, and arm raises) to reinforce social presence and promote emotional connection.

3.2 Gamified Rehabilitation Exercises

The gamified rehabilitation exercise platform was developed in collaboration with clinical rehabilitation specialists and according to cardiac rehabilitation guidelines based on exercise in the early-phase [38]. The specialists specified the ranges of motion and the frequency of exercise (see Tab. 1) to meet the safety standards for the inpatient rehabilitation phase. These parameters ensured that physical exertion remained within a safe metabolic equivalent of task range of approximately 2 to 4. The platform integrated game elements [34], including narrative structure, levels and points.

The platform uses a progressive narrative titled “Gradual Healing”, in which the patient protects and restores a health crystal. This metaphor reframes repetitive exercise movements as meaningful game tasks in five different stages, as depicted in Fig. 3. The stages are:

- Stage 1 (Intrusion Defense): The patient performs vertical arm raises to control the vertical position of a pink companion robot, aiming and firing projectiles to neutralize incoming enemies and

protect the health crystal.

- Stage 2 (Element Elimination): After the attack, the residual detrimental elements adhere to the health crystal. The patient performs circular shoulder rotations to guide the robot in eliminating these elements sequentially.
- Stage 3 (Energy Tide): With the detrimental elements removed, the health crystal requires repair. The patient performs ankle pumps to direct the robot to deliver a continuous infusion of healing energy into the crystal.
- Stage 4 (Energization Balance): Infusion of energy induces internal instability within the health crystal. The patient performs alternating leg raises to stabilize the energy and restore equilibrium.
- Stage 5 (Health Resonance): Once balanced and fully energized, the health crystal emits health waves. The patient performs glute bridges to synchronize and absorb these waves.

A point-based scoring mechanism was implemented to provide extrinsic rewards. Feedback was triggered only upon the successful execution of a prescribed movement. Tab. 1 specifies the movement ranges required to earn points and the repetition counts needed to advance to the next stage. Meeting these thresholds triggered three concurrent feedback modalities: a visual particle effect with an on-screen score increment labeled “POINT”, an auditory cue and a haptic vibration (150 Hz for 500 ms) delivered through the VR controllers. Participants

Table 1: Summary of the target movement range to trigger game points and feedback, and frequency required to advance to the next stage.

Stage	Exercise	Movement Range	Frequency
1	Arm	60°	25
2	Shoulder	30 cm	25
3	Ankle Pump	45°	25
4	Leg Raising	45°	25
5	Glute Bridge	10 cm	10

could skip any stage; however, points were awarded only for repetitions that satisfied the predefined criteria.

This design excludes the implementation of countdown timers for older adults, allowing self-paced activity that supports emotion-focused goals and minimizes negative experiences, such as time pressure, during challenges [45]. This approach supports a patient-centered and safe rehabilitation process. The design implementation materials are available at [57].

4 EXPERIMENT DESIGN

4.1 Participants

The study involved 30 hospitalized patients diagnosed with AMI (19 males, 11 females), aged 47 to 78 years ($\mu_{age} = 60.7$, $\sigma_{age} = 9.6$), at the Affiliated First Hospital of Ningbo University. No participant had previous experience with IVR gaming. The hospital rehabilitation specialists verified that all subjects met the following inclusion criteria: (1) voluntary participation and capacity to provide signed written informed consent; (2) adequate visual and auditory function for interaction with the IVR environment; and (3) absence of medications known to alter heart rate (HR) responses during exercise. Exclusion criteria were as follows: (1) incomplete revascularization after percutaneous coronary intervention for AMI; (2) pre-existing lower back or limb pain; (3) presence of indwelling tubes (e.g., femoral vein hemodialysis catheters, temporary pacemakers); (4) muscle strength test score below grade five according to the hospital's scale; (5) dyspnea at rest accompanied by pulmonary crackles; (6) chest pain, chest tightness or arrhythmia within the previous eight hours; (7) myocardial injury (troponin) or cardiac function (brain natriuretic peptide) that is beyond clinical standards; (8) left ventricular ejection fraction less than 40%; and (9) severe cognitive impairment that could influence the comprehension of instructions. Physiological and user experience data were collected without personally identifiable information to preserve participant anonymity. The study protocol was reviewed and approved by the Code of Practice for Research Ethics Committees of the University of Nottingham Ningbo China [55], under approval number FOSE-202425-060.

4.2 Experiment Setup

The IVR-assisted rehabilitation game was developed using Unity 3D Editor v6000.0.43f1 [54] and installed on a PICO 4 Ultra IVR headset [41]. The headset offers a 90 Hz refresh rate, 2160×2160 pixels per eye resolution, and a 105° field of view. A laptop equipped with an NVIDIA RTX 2000 Ada Generation GPU, an Intel i7-13800H CPU, and 32 GB RAM was used to mirror IVR content on its monitor for observation. Fig. 4 illustrates the hospital experimental setup, showing a patient wearing the VR headset and associated equipment. Two PICO VR controllers (6 DoF) capture hand movements. Three PICO motion trackers (6 DoF) [40], positioned on the waist and each foot, tracked the movements of the lower limbs and waist with a latency of 15.6 ± 2.6 ms. A Polar Verity Sense armband [43] collected HR data from the arm. The 3D camera system of the game was placed semi-reclined to align with a patient lying on a bed inclined at approximately 30° to 45° . Nurses re-adjusted the angle of inclination of the bed to 0° to obtain an optimal position during glute bridge exercise (stage 5).

Fig. 5 depicts the experimental procedure. The experiment was carried out in the hospital's coronary care unit (CCU). It started with a 10-minute briefing that described the study's objectives. The participants then signed written informed consent forms once they agreed to

participate. A researcher and a nurse assisted participants in wearing the VR headset and equipment, which took approximately 3 minutes, followed by a 5-minute tutorial to familiarize participants with the operations and interactions of the game. Participants retained the right to discontinue at any time if they experienced discomfort.

Throughout the session, the researcher, a rehabilitation specialist, and a nurse were present to ensure safety. The nurse continuously monitored the patient's vital signs using the ward's standard vital signs monitor, under the specialist's supervision. The safety protocol required immediate termination of the session if the patient's HR exceeded the clinical safety threshold of 20 beats per minute (BPM) above the resting HR. Given these safety measures and strict pre-participation screening, no adverse events were reported during the study.

This study used a within-subjects design which required each patient to complete two games sessions on separate days: one with VPM integration and one baseline session without it. Each session lasted approximately 15 minutes on average, with slight variations depending on individual pace. To control order effects, patients were assigned to two groups using a rotation method: Group 1 played the VPM-integrated session first; Group 2 started with the baseline session. Participants received a 5- to 10-minute break after the game. They then completed self-rating questionnaires that took about 10 minutes. On the second day, in addition to the similar procedure in the first day, participants also participated in a 10-minute semi-structured interview to provide in-depth feedback.

4.3 Measurements

Both baseline and VPM conditions offered identical five-stage gamified rehabilitation exercises, featuring matching game tasks, movement targets, and basic visual, auditory, and haptic feedback. The primary distinction lies in social support: the VPM-integrated condition included three types of social support (see Sec. 3.1), while the baseline condition lacked any of these features. Although both conditions provided identical motion guidance demonstrations, the VPM condition portrayed them as the VPM's own successful rehabilitation experiences, while the baseline condition presented them solely as instructional content without peer representation.

This study used the Intrinsic Motivation Inventory (IMI) [29], Game Experience Questionnaire – Social Presence Module (GEQ-SPM) [19], Virtual Reality Sickness Questionnaire (VRSQ) [21], in-game performance metrics and physical exertion data as quantitative measures, along with semi-structured interview feedback as qualitative data. The paired t-tests were used to compare baseline and VPM-integrated conditions.

To explore possible age-related differences in response to the intervention, this study conducted a post hoc subgroup analysis. Participants were stratified into two age groups: middle-aged (< 60 years old, $n = 14$) and older (≥ 60 years old, $n = 16$). Within each subgroup, paired-samples t-tests were performed to evaluate the specific impact of the VPM between age groups.

4.3.1 IMI

The IMI assesses key dimensions of subjective experience and intrinsic motivation on a 7-point Likert scale (1: "Not At All True" to 7: "Very True"). This instrument directly addresses RQ1 by quantifying pressure/tension to assess stress mitigation and interest/enjoyment to measure motivational impact. The dimensions of perceived competence and perceived choice further indicate the influence of VPM on perceptions of mastery and autonomy.

4.3.2 GEQ-SPM

The GEQ-SPM assessed the VPM's ability to promote companionship and social support, directly related to the behavioral involvement component in RQ1. This 5-point Likert scale instrument (0: "Not At All" to 4: "Extremely") measures three core dimensions: empathy (connection and shared experience), negative feelings (potential stress from virtual presence) and behavioral involvement (active engagement in a shared VPM-integrated activity).

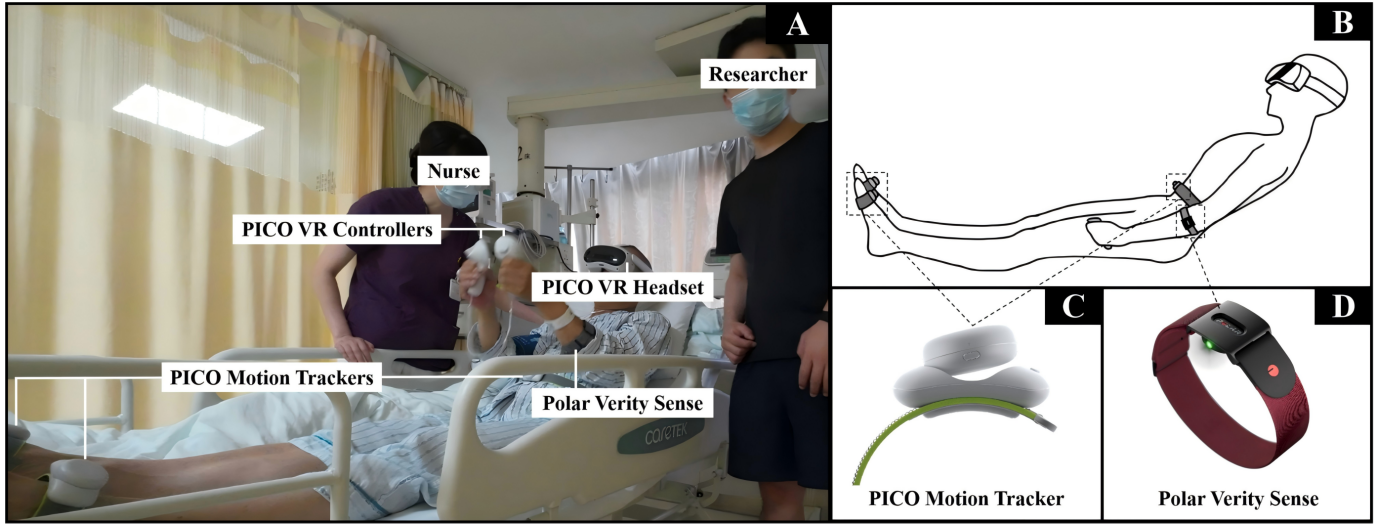


Fig. 4: Overview of the experimental setup. (A) A patient is equipped with a PICO 4 Ultra VR headset, two PICO VR controllers, three PICO motion trackers, and a Polar Verity Sense armband. A nurse and researcher are present to observe the session. (B) Placement of wearable sensors on the patient body. (C) Detailed view of a PICO Motion Tracker. (D) Detailed view of a Polar Verity Sense.

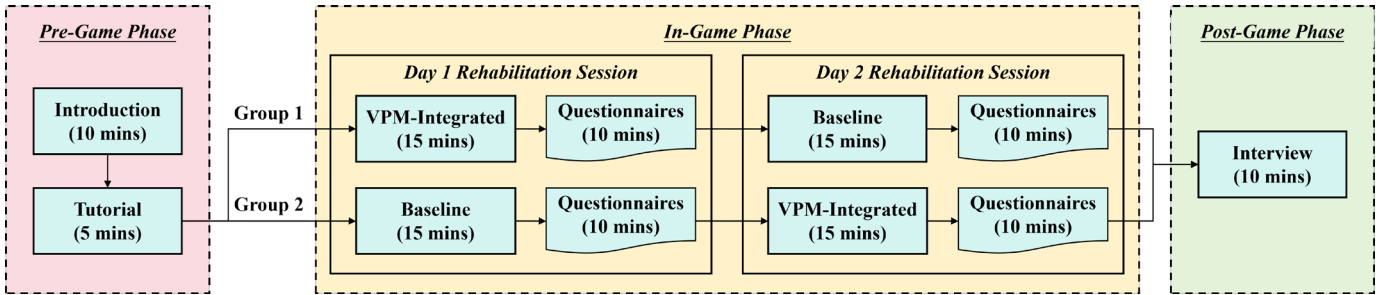


Fig. 5: Illustration of the experimental procedure including the pre-game, in-game, and post-game phases, used a within-subjects counterbalancing technique.

4.3.3 VRSQ

The VRSQ measured symptoms such as general discomfort, fatigue, eye strain, difficulty focusing, headache, head fullness, blurred vision, dizziness, and vertigo on a 4-point Likert scale (0: “Not At All” to 3: “Very”). This measure controlled for technological discomfort, ensuring that observed differences in other metrics derived from the VPM-integrated intervention were not cybersickness.

4.3.4 In-Game Performance Metrics

Two objective in-game performance metrics were recorded as indicators of engagement and task efficiency. Each stage completion rate was calculated as the average of correct movements (meeting the target range) divided by the target movement frequency. Then, the overall mean completion rate was calculated as the average of the five stage completion rates. The completion time measured the total duration from the initiation of the game to the completion within a single session, providing an overall assessment of the sustainability of the engagement of the participants. The participants retained the autonomy to skip stages.

4.3.5 Physical Exertion

HR, expressed as mean BPM, was recorded throughout the experiment using a Polar Verity Sense. This provided a physiological indicator of exercise intensity and a potential objective correlate of psychological engagement and emotional state. HR data served to physiologically corroborate the self-reported pressure/tension scores from the IMI.

4.3.6 Semi-Structured Interview

Semi-structured interviews explored subjective experiences of participants with the IVR-assisted rehabilitation game through five open-ended questions that covered three aspects (see Tab. 2). The interviews were conducted in Chinese to facilitate natural communication. A general inductive approach [51] was adopted, involving systematic data preparation, transcript reading, iterative coding, and thematic categorization to analyze the recorded responses.

5 RESULTS

Tab. 3 presents descriptive statistics and paired t-test results for various quantitative measures comparing the VPM-integrated and baseline conditions of the custom IVR-assisted rehabilitation game.

5.1 IMI Results

The analysis of the IMI revealed varied patterns in its four dimensions. The most prominent finding occurred in the pressure/tension dimension, which showed significantly lower scores in the VPM-integrated condition than in the baseline condition ($t = -2.697, p < .05, d = -.49$). This moderate effect size indicates that the social support of the VPM reduces stress during rehabilitation exercises. The scores for interest/enjoyment were marginally higher in the VPM-integrated condition, although this difference only approached statistical significance ($t = 1.916, p = .065, d = .35$). A similar marginal and non-significant trend was observed that favors the VPM condition in the perceived competence dimension ($t = 1.836, p = .077, d = .34$). The perceived choice dimension did not show significant differences between conditions ($t = -1.021, p = .316, d = -.19$), indicating that the VPM did

Table 2: List of interview questions covering three aspects: motivation, perceived social support, and overall user experience.

No.	Question
Motivation:	
1	Could you describe your experience with both exercise sessions? What, if anything, stood out to you about it?
Perceived Social Support:	
2	How would you describe your experiences with the virtual character (the mentor) during the session?
3	Can you tell me about a specific moment involving the character that you remember clearly?
Overall User Experience:	
4	Thinking about the entire session, what are your overall impressions?
5	What did you like the most? What did you like the least?
6	How did this experience compare to your expectations of rehabilitation?
7	How do you feel about the idea of using technology like this as part of rehabilitation in the future?

not affect the autonomy of the participants.

The subgroup analysis indicated that the observed reduction in pressure/tension was driven by the middle-aged group, which showed a statistically significant decrease ($t = -2.18, p < .05, d = -.58$). In contrast, the older group exhibited a non-significant reduction ($t = -1.30, p = .213, d = -.33$). In other IMI dimensions (interest/enjoyment, perceived competence, and perceived choice), no statistically significant differences were detected within the group between the VPM-integrated and baseline conditions in either age subgroup.

5.2 GEQ-SPM Results

The social presence analysis indicated a significant difference in behavioral involvement, with scores significantly exceeding those under the baseline condition ($t = 2.526, p < .05, d = .46$). This moderate effect size suggests that VPM-integrated condition facilitated greater active engagement during exercises. Empathy scores exhibited a non-significant trend towards higher values in the VPM-integrated condition ($t = 1.449, p = .158, d = .26$), suggesting a modest sense of connection with the virtual companion. No significant differences in negative feelings were observed between the conditions ($t = .540, p = .593, d = .10$). Low scores in both conditions indicate that neither the presence nor the absence of VPM elicited negative feelings. The subgroup analysis revealed no statistically significant differences in any GEQ-SPM dimension between the VPM-integrated and baseline conditions in either age subgroup.

5.3 VRSQ Results

The symptoms of cybersickness remained minimal in both conditions, with no statistically significant differences. Oculomotor symptoms did not show significant differences between the VPM-integrated and baseline conditions ($t = .578, p = .568, d = .11$). Similarly, the symptoms of disorientation did not show a significant difference ($t = .291, p = .773, d = 0.05$). These negligible levels indicate that the observed differences in other measures can be attributed to the VPM-integrated intervention rather than to technology-related discomfort. The subgroup analysis indicated that neither age group exhibited statistically significant differences between-conditions in oculomotor or disorientation symptoms.

5.4 In-Game Performance Results

Analysis of in-game performance metrics indicated that there were no statistically significant differences between the conditions. The completion rates remained high under both conditions with no significant differences ($t = -1.174, p = .250, d = -.21$). The completion times

were also virtually identical between conditions ($t = .053, p = .958, d = .01$). These null results suggest that the social support of the VPM did not compromise the competence to carry out the task. No participants withdraw from the experiments. The subgroup analysis revealed no statistically significant differences in the completion rate or completion time between conditions in the middle-aged or older subgroup.

5.5 Physical Exertion Results

The HR data did not show significant differences between the conditions ($t = .036, p = .970, d = .01$). The nearly identical mean values and negligible effect size suggest that the VPM did not induce further physiological stress. Stable HR responses suggest that participants maintained comparable levels of physical exertion irrespective of social support elements. The subgroup analysis of measured HR did not show statistically significant differences between the VPM-integrated and baseline conditions in either age subgroup.

5.6 Interview Results

Inductive thematic analysis of qualitative interview data revealed three primary themes: peer support and connection; acceptance of the IVR intervention; and recommendations for improvement. The categorization of the responses indicated that 56.7% of the participants ($n = 17$) valued the social support of the VPM, while a minority ($n = 3$) preferred real human interaction.

Game acceptance was high: 76.7% of the participants ($n = 23$) expressed willingness to continue using the IVR-assisted rehabilitation game in the future, in the context of a comparison with their usual (traditional) rehabilitation routine. In addition, 30% ($n = 9$) reported that the game was easy to understand. The constructive feedback from 36.7% of the participants ($n = 11$) reflected the desire for greater variety and challenge of the content, while 16.7% ($n = 5$) reported unsatisfactory technical limitations in motion tracking, specifically experiencing micro-dropouts manifested as less responsive tracking of the wearable motion trackers. Moreover, 6.67% of participants ($n = 2$) expressed a desire to repeat the IVR session, indicating that they could continue the exercises at higher intensity. A participant briefly paused the session (approximately one minute) during both the shoulder exercise stage and the glute bridge stage due to fatigue, then resumed and completed the game.

6 DISCUSSION

RQ1: Does the integration of a VPM into the feedback mechanism reduce short-term pressure and tension while increasing behavioral involvement among middle-aged and older AMI patients in IVR-assisted rehabilitation?

This study demonstrates that the integration of VPM improves motivation and behavioral involvement in IVR-assisted rehabilitation among middle-aged and older patients with AMI. The most significant result was a reduction in pressure/tension scores in the VPM-integrated condition compared to baseline, indicating a psychological benefit in line with social support theory. In contrast to traditional virtual coaches or instructors [17], the VPM is designed as a peer experiencing similar rehabilitation challenges, thus creating a more supportive and less stressful environment. This finding is consistent with research highlighting the therapeutic value of connecting patients with comparable health trajectories [12] and highlights how the built-in peer support mechanism of the VPM reinforces this therapeutic bond.

Specifically, informational support, provided through shared medical narratives, clarified the rehabilitation process and likely reduced pressure by offering relatable and understandable frameworks [12]. Instrumental support, provided through peer movement demonstrations, offered technical guidance and proof of achievability, which promoted self-efficacy and increased behavioral involvement, in line with the findings of [49]. Emotional support, provided through verbal affirmations and positive gestures, validated patient efforts, reduced tension, and maintained motivation, correlating with the cardiac rehabilitation

Table 3: Paired t-test results comparing VPM and baseline conditions for different quantitative measures.

Measurement	Dimension	Conditions ($\mu \pm \sigma$)		t	p	Cohen's d
		VPM-Integrated	Baseline			
IMI	Interest/Enjoyment	5.41 $\pm$ 1.09	5.26 $\pm$ 0.93	1.916	.065	0.35
	Perceived Competence	5.45 $\pm$ 0.74	5.31 $\pm$ 0.73	1.836	.077	0.34
	Perceived Choice	4.41 $\pm$ 0.76	4.47 $\pm$ 0.65	-1.021	.316	-0.19
	Pressure/Tension	2.12 $\pm$ 0.56	2.39 $\pm$ 0.48	-2.697	< .05*	-0.49
GEQ-SPM	Empathy	1.49 $\pm$ 0.35	1.36 $\pm$ 0.47	1.449	.158	0.26
	Negative Feelings	0.97 $\pm$ 0.36	0.91 $\pm$ 0.56	0.540	.593	0.10
	Behavioral Involvement	2.36 $\pm$ 0.53	2.03 $\pm$ 0.47	2.526	< .05*	0.46
VRSQ	Oculomotor	0.83 $\pm$ 0.42	0.75 $\pm$ 0.32	0.578	.568	0.11
	Disorientation	0.76 $\pm$ 0.57	0.73 $\pm$ 0.44	0.291	.773	0.05
In-Game Performance Metrics	Completion Rate (%)	98.19 $\pm$ 4.20	98.72 $\pm$ 3.64	-1.174	.250	-0.21
	Completion Time (min)	13.79 $\pm$ 1.83	13.78 $\pm$ 1.96	0.053	.958	0.01
Physical Exertion	HR (BPM)	70.16 $\pm$ 6.65	70.13 $\pm$ 6.47	0.036	.970	0.01

study [37]. In general, these three types of support function synergistically: informational support establishes understanding, instrumental support builds competence, and emotional support maintains engagement throughout the rehabilitation process. However, these findings reflect within-session effects and do not necessarily indicate long-term adherence or sustained motivational benefits.

This psychological benefit is corroborated by interview data, in which 56.7% of the participants ($n = 17$) reported receiving encouragement and social support during the VPM-integrated sessions. Qualitative responses also indicated that participants perceived the VPM as different from traditional authority figures, supported by the statement of participant P3:

“Unlike a doctor or trainer, it felt like having a friend who understood what I was going through.”

This distinction reinforces the conceptualization of the avatar as a peer. Furthermore, participants P3, P7, P11, P20, and P29 consistently noted that VPM alleviated feelings of loneliness, a particularly significant concern for middle-aged and older patients with AMI who undergo prolonged bed rest.

In contrast, a minority of participants (P6, P26, P27) reported unfavorable experiences. These participants, including two women aged 49 and 68 and one man aged 70, did not share age- or gender-related characteristics predictive of dissatisfaction. Instead, their feedback indicated a common resistance to virtual environments itself, reflected in a general preference for physical reality over virtual settings or humanoid avatars. This finding suggests that, for a subset of participants, the primary barrier is not the specific design of the VPM, but the virtual modality itself. For instance, participant P6 noted:

“I cannot feel any social support from any representation of virtual avatars; what I want is only the company of real humans.”

Although no participants reported an uncanny valley effect associated with VPM, realism-related discomfort remains a critical consideration when designing avatars for older adults. The study showed that greater realism does not inherently increase trust; instead, participants frequently preferred intermediate stylization [6]. Moreover, behavioral inconsistencies (e.g., lag in facial motion, unnatural movements, or mismatched textures) can elicit uncanny impressions and reduce the willingness to trust. This points to a broader implication for rehabilitation: in vulnerable healthcare settings, trust is shaped not only by the human likeness of an avatar, but also by whether its appearance, motion, and interaction style collectively form a coherent and believable entity.

Marginal improvements in interest/enjoyment and perceived competence, although not statistically significant, suggest a trend toward increased short-term intrinsic motivation under the VPM-integrated

condition. A significant increase in behavioral involvement, as captured by the GEQ-SPM, provides evidence that VPM promotes more active participation in the rehabilitation. Even though existing studies reported favorable patient attitudes towards non-humanoid virtual companions [31], they lack quantitative measures of behavioral involvement or social presence. These findings demonstrate that a VPM structured around shared experiences can enhance behavioral engagement beyond mere satisfaction.

Consistently high in-game completion rates, efficient completion times, and stable HR responses in both conditions indicate that the difficulty of the game was well calibrated.

RQ2: What key factors influence the acceptance and experience of IVR-assisted rehabilitation among middle-aged and older AMI patients?

The acceptance rate of 76.7% for future use of the game suggests potential for IVR-assisted rehabilitation in this demographic group. This rate appears to be favorable compared to the general adoption rates among middle-aged and older adults [59]. The design emphasis on passive immersion over active exploration may have contributed to this acceptance, as the game kept patients seated in bed within immersive environments and limited interactions to guided therapeutic movements and social engagement with the VPM. 30% of the participants ($n = 9$) commented positively on the simplicity and focus of the game, as illustrated by participants P10 and P17:

“I appreciated that I could do all the exercises while lying on the bed. It felt safe and appropriate for my current condition.”

“This game was straightforward and did not require a lot of complicated interaction, so I could focus on completing the game tasks.”

These findings are consistent with studies indicating that the resistance of older adults to IVR can arise from the cognitive burden associated with proactive engagement [46], suggesting that guided, purpose-driven activities could help address barriers to IVR adoption.

The clinicians involved in the study noted the potential advantages of IVR-assisted rehabilitation over traditional approaches in the hospital setting. The immersive environment presents movement guidance with greater visual clarity, potentially reducing ambiguity about correct execution. In addition, the IVR game provides immediate repetition-by-repetition feedback, whereas traditional rehabilitation often relies on intermittent supervision.

The minimal cybersickness scores on the VRSQ indicate that the bed-based IVR game effectively addresses the physical limitations and comfort needs of post-AMI patients. These scores are lower than those

reported for comparable IVR systems, which frequently exhibit moderate to high levels of cybersickness [28]. By allowing patients to remain supine, the bed-based design appears to mitigate motion-induced discomfort while preserving immersive engagement, and no participant reported dizziness during interviews. This outcome alleviates concerns about the suitability of IVR for middle-aged and older adults who require rehabilitation [25].

The high overall completion rates confirm that rehabilitation exercises are properly calibrated for this cohort of patients; however, the somewhat lower completion rate of 93.5% for glute bridge exercise highlights inherent challenges, possibly due to supine positioning demands or pelvic elevation biomechanics. This pattern persisted in both the VPM-integrated and the baseline conditions, indicating that the difficulty is intrinsic to the exercise itself rather than resulting from the VPM intervention.

The feedback from the interviews highlights critical areas for game improvement: 36.7% of the participants ($n = 11$) cited insufficient variation in the narratives and tasks and a lack of challenge, while 16.7% ($n = 5$) reported technical limitations, for example, participants P15 and P20 noted that:

"Sometimes, the motion tracking seemed a bit delayed."

"Motion trackers sometimes did not work properly."

These findings further indicate that technical limitations, such as delayed or inconsistent motion tracking, adversely influence the overall experience of middle-aged and older adults with the IVR-assisted rehabilitation game, potentially affecting their engagement and therapeutic results.

6.1 Limitations and Future Work

This study presents several limitations. The limited sample size ($n = 30$) reduces statistical power, restricting the generalization of the results. Furthermore, although the hospital setting ensures clinical safety, it may not accurately represent cardiac rehabilitation at home, affecting the ecological validity of the acceptance and experiences reported by middle-aged and older adults. The limitations of IVR motion tracking highlight the need for reliable tracking in clinical settings; inaccuracies can interfere with therapy and patient experience.

From a technical point of view, the reported tracking issues were traced to intermittent latency spikes. The tracking system relied on frame-dependent synchronous polling; consequently, fluctuations in visual rendering frame rates could delay sensor data retrieval. In addition, the tracking algorithm used sequential device assignment without persistent binding to unique identifiers, which could introduce momentary reassignment delays when sensor signals fluctuated. The game also applied fixed motion thresholds across all exercises without adjusting for participant anthropometrics (e.g., stature and mass). As a result, the current study was unable to empirically assess how these factors affected perceived difficulty, including whether fixed ranges imposed disproportionate biomechanical demands between individuals.

Future work should consider larger and more diverse samples to enhance statistical power and enhance external validity in various rehabilitation contexts. This study laid the groundwork by implementing a fully scripted and standardized VPM framework. Building on this foundation, future VPM implementations will transition from static scripting to an adaptive architecture. The planned capabilities include rule-based adaptations, for example, dynamically adjusting exercise pace based on a detected fatigue proxy or error patterns, and lightweight personalization features. In addition, a voice-only support mode is expected to benefit users who are sensitive to humanoid avatars. Subsequent iterations will also address biomechanical variability through dynamic calibration: instead of fixed values, ranges of motion will be scaled relative to individual anthropometry. The short duration of the intervention precludes definitive conclusions about long-term engagement and the durability of therapeutic benefits. Because the benefits of social support are often cumulative, extended longitudinal studies are needed to determine whether VPM improves sustained adherence and promotes long-term behavior change over the course of a cardiac rehabilitation program.

Future research will also investigate how the appearance of the virtual mentor affects clinical trust and user experience by comparing various representations (e.g., human-like, robot-like, animal-like, abstract and voice-only). Although human-like forms can increase social presence, they also risk realism-mismatch effects if responsiveness is constrained. Controlled evaluations are needed to characterize trade-offs between perceived support, comfort, and cognitive load between avatar types. These findings will inform the selection of an optimal mentor appearance for older cardiac patients and the development of a clinical protocol in which patients resistant to humanoid avatars can select for non-humanoid or voice-only alternatives. In addition, future work will address tracking latency by implementing asynchronous data polling and persistent device-identifier binding to improve stability independent of rendering frame rates.

7 CONCLUSION

This study introduces an IVR-assisted cardiac rehabilitation game that integrates a VPM grounded in social support theory and employs a mixed-methods design to evaluate its impact on middle-aged and older patients with AMI. Quantitative results demonstrate that VPM-integrated condition significantly reduces perceived pressure and tension while increasing behavioral involvement during rehabilitation, while qualitative findings indicate that the VPM promotes companionship and peer support, mitigating loneliness during hospitalization. Furthermore, the intervention achieved an acceptance rate of 76.7% and elicited minimal cybersickness, highlighting key factors influencing the adoption and user experience in this demographic. These insights offer valuable guidance for the design of effective digital health interventions, and future work will conduct longitudinal studies of AI-driven VPMs in diverse rehabilitation contexts to establish a broader generalizability.

ACKNOWLEDGMENTS

This work was supported by the UNNC Education Foundation through the Li Dak Sum Innovation Fellowship under Grant LDS202307.

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